

LUCIDPREP

Grade 10 Geometry

U.S. Standards-Aligned Course Syllabus

GEOMETRY

Geometry · Common Core

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

The U.S. has no single national curriculum, but Geometry is the standard Grade 10 course in the most common sequence (Algebra 1 → Geometry → Algebra 2). Some states and schools use an Integrated Mathematics pathway (Math I/II/III) that distributes the same standards differently. This syllabus uses the Common Core high-school Geometry category as the primary reference and cross-checks against California and Texas TEKS.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

GEOMETRY

Grade 10 Geometry aligned to
the Common Core State
Standards

This 36-week course is designed around **the Common Core high-school Geometry category: Congruence (G-CO), Similarity & Right-Triangle Trigonometry (G-SRT), Circles (G-C), Expressing Geometric Properties with Equations (G-GPE), Geometric Measurement & Dimension (G-GMD), Modeling with Geometry (G-MG) and Conditional Probability (S-CP)**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 10
Course	Geometry · Common Core
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Reason with definitions, constructions and rigid transformations.
- Prove and apply theorems about lines, angles, triangles and quadrilaterals.
- Use similarity and right-triangle trigonometry to solve problems.
- Analyze circles and geometric figures on the coordinate plane.
- Compute measurements and model real-world situations geometrically.

GEOMETRY

Module 1 — Foundations, Transformations & Congruence

Week	Topics	Students will be able to	Standards
1 — Building Blocks & Constructions	Points/lines/planes, definitions, compass-and-straightedge constructions	Use precise definitions and perform basic constructions	G-CO.1/12
2 — Rigid Transformations	Translations, reflections, rotations; properties	Describe and perform rigid motions	G-CO.2/4/5
3 — Symmetry & Mapping Figures	Symmetry; carrying figures onto themselves	Analyze symmetry via transformations	G-CO.3
4 — Congruence via Transformations	Definition of congruence; corresponding parts	Establish congruence with rigid motions	G-CO.6
5 — Triangle Congruence Criteria	ASA, SAS, SSS from transformations	Prove triangle congruence	G-CO.7/8
6 — Proving Line & Angle Theorems	Vertical angles; parallel lines and transversals	Prove line and angle theorems	G-CO.9
7 — Proving Triangle Theorems	Angle sum, isosceles, midsegment	Prove triangle theorems	G-CO.10
8 — Proving Quadrilateral Theorems	Parallelograms and their properties	Prove parallelogram theorems	G-CO.11
9 — Module 1 Assessment	Cumulative review of congruence	Demonstrate mastery of Module 1	G-CO

Module delivery notes. Center reasoning and proof: build from transformations and constructions to formal argument. Connect synthetic and coordinate approaches, and anchor measurement in real modeling.

GEOMETRY

Module 2 — Similarity & Right-Triangle Trigonometry

Week	Topics	Students will be able to	Standards
1 — Dilations & Scale	Dilations, scale factor, center	Perform and analyze dilations	G-SRT.1
2 — Similarity Transformations	Definition of similarity; AA criterion	Establish similarity	G-SRT.2/3
3 — Proving Similarity Theorems	Side-splitter; similar triangles	Prove and use similarity theorems	G-SRT.4/5
4 — Right-Triangle Similarity	Geometric mean; altitude relationships	Apply right-triangle similarity	G-SRT.4
5 — Trigonometric Ratios	Sine, cosine, tangent from similarity	Define and compute trig ratios	G-SRT.6/7
6 — Solving Right Triangles	Finding sides and angles; inverse trig	Solve right triangles	G-SRT.8
7 — Applications of Trigonometry	Angles of elevation/depression; modeling	Model with right-triangle trigonometry	G-SRT.8
8 — Laws of Sines & Cosines	Non-right triangles	Apply the laws of sines and cosines	G-SRT.10/11
9 — Module 2 Assessment	Cumulative review of similarity and trig	Demonstrate mastery of Module 2	G-SRT

Module delivery notes. Center reasoning and proof: build from transformations and constructions to formal argument. Connect synthetic and coordinate approaches, and anchor measurement in real modeling.

GEOMETRY

Module 3 — Circles & Coordinate Geometry

Week	Topics	Students will be able to	Standards
1 — Circle Angles & Arcs	Central and inscribed angles; arcs	Relate angles and arcs	G-C.2
2 — Chords, Tangents & Secants	Segment and tangent relationships	Apply chord/tangent/secant relationships	G-C.2
3 — Arc Length & Sectors	Radian intro; arc length; sector area	Compute arc length and sector area	G-C.5
4 — Equations of Circles	Deriving from the Pythagorean theorem; completing the square	Write and use circle equations	G-GPE.1
5 — Coordinate Proofs	Proving theorems algebraically	Prove theorems using coordinates	G-GPE.4
6 — Lines on the Plane	Slope of parallel/perpendicular; partitioning segments	Use slope criteria and partition segments	G-GPE.5/6
7 — Perimeter & Area on the Plane	Computing with coordinates	Compute perimeter and area via coordinates	G-GPE.7
8 — Parabolas	Focus-directrix definition	Derive a parabola from focus and directrix	G-GPE.2
9 — Module 3 Assessment	Cumulative review of circles and coordinates	Demonstrate mastery of Module 3	G-C/G-GPE

Module delivery notes. Center reasoning and proof: build from transformations and constructions to formal argument. Connect synthetic and coordinate approaches, and anchor measurement in real modeling.

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Module 4 — Measurement, Modeling & Probability

Week	Topics	Students will be able to	Standards
1 — Area & Volume Formulas	Deriving formulas; Cavalieri's principle	Explain and use area/volume formulas	G-GMD.1
2 — Volume of Solids	Prisms, cylinders, cones, spheres	Compute volumes	G-GMD.3
3 — 2D / 3D Relationships	Cross-sections; rotations of 2D figures	Relate two- and three-dimensional figures	G-GMD.4
4 — Modeling with Geometry	Density; design; real objects	Model situations with geometry	G-MG.1/2/3
5 — Probability & Sample Spaces	Events, sample spaces, set operations	Describe events and sample spaces	S-CP.1
6 — Conditional Probability	Independence; conditional probability	Compute and interpret conditional probability	S-CP.2/3
7 — Two-Way Tables & Independence	Interpreting; testing independence	Use two-way tables	S-CP.4/5
8 — Probability Rules	Addition and multiplication rules	Apply probability rules	S-CP.6/7
9 — Final Assessment	Cumulative final across all modules	Demonstrate mastery of the full course	G-CO-S-CP

Module delivery notes. Center reasoning and proof: build from transformations and constructions to formal argument. Connect synthetic and coordinate approaches, and anchor measurement in real modeling.

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Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	CCSS: G-CO, G-SRT, G-C, G-GPE, G-GMD, G-MG, S-CP
Mathematical Practices	Make sense of problems and persevere; reason abstractly and quantitatively; construct viable arguments and critique reasoning; model with mathematics; use tools strategically; attend to precision; use structure and regularity.
Recurring Concepts	Multiple representations; equivalence; proportional reasoning; structure; modeling; real-world application.
State mapping	Common Core adopted by the large majority of states; Texas TEKS and other state frameworks cover the same grade-level content under different codes.

Content Production Guidance

Author each week as 5 warm-up items, a concept lecture with worked examples and multiple representations, 2 curated videos, an AI presentation with models and strategies, 2 homework sets (5 items each), and a 40-question bank spanning fluency, application and reasoning. Tag every item to its Common Core code.

Curriculum Basis & Source Notes

This Grade 10 mathematics syllabus was generated from the U.S. Common Core State Standards for Mathematics. Content and pacing reflect Geometry, the typical Grade 10 course in the most common U.S. sequence (Algebra 1 → Geometry → Algebra 2).

Primary reference	Common Core State Standards for Mathematics — high-school Geometry conceptual category (G-CO, G-SRT, G-C, G-GPE, G-GMD, G-MG) and the Standards for Mathematical Practice.
State cross-check	California Common Core and integrated-pathway (Math I/II/III) standards.
State cross-check	Texas Essential Knowledge and Skills (TEKS) for Geometry.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module secondary structure with 70% mastery.

Recommended next production step. Commission item banks per week (40 assessment items plus warm-ups and homework), each tagged to its Common Core code, then load into the LucidPrep authoring pipeline.